PS2_oaxaca_sln

September 10, 2025

Note: Write your code in the code cells, and your responses in markdown. Run the entire script and display the outputs of your code.

Due: **11:59PM Central Time on Friday, 09/26**. Upload both your code (.ipynb) and responses (html or pdf) to Canvas by then.

[2]:
```
# Packages you might need: (pip install ... if you don't have them)
import pandas as pd
import os
print(os.getcwd())
import statsmodels.formula.api as smf
import numpy as np
```

```
c:\Users\wuhao\Dropbox\UW695\problem_sets\PS2
```

1 Oaxaca Decomposition

The goal of this problem set is to extend the analysis of public-private pay gaps in Lecture 3 to male workers. The file pay.csv contains 45,404 observations on male workers, age 30-50, with education of 12 or more years, in the March 2012 and 2013 CPS files. The variables are: - govt=1 if government worker - logwage=log hourly wage based on earnings, weeks worked, and average hours per week last year - educ = years of education - hs = 1 if high school grad (12 years of schooling) - somecoll=1 if education is 13-14 years. - college = 1 if BA (16 years of education) - ma = 1 if masters degree (18 years of schooling) - phd = 1 if phd or LLD or medical degree (20 years of schooling) - age= age in years (range 30-50)

NOTE: note that "somecoll" combines 2 levels of education, 13 and 14 years (13 years are people with some college but no degree, 14 years are people with an AA degree from a community college).

Please create 2 new variables: educ13=1[educ=13] and educ14=1[educ=14] so you will have a total of 6 possible categories for education.

[3]:
```
# Load dataset (using pandas "pd")
pay = pd.read_csv("pay.csv") # add your own directory if necessary

# create 2 new variables: educ13, educ14
pay['educ13'] = 0+(pay['educ']==13)
pay['educ14'] = 0+(pay['educ']==14)
print(pay.head(2))
```

```
# 6 educ categories (mutually exclusive): phd, ma, college, educ14, educ13, hs
assert (1==pay[['hs','educ13','educ14','college','ma','phd']].sum(axis=1)).all()
# define a variable for 6 education levels:
pay['educ_level'] = 1*(pay['hs']==1) + 2*(pay['educ13']==1) +␣
↪3*(pay['educ14']==1) + 4 * (pay['college']==1) +5* (pay['ma']==1) + 6 *␣
↪(pay['phd']==1)
print(pay['educ_level'].value_counts(dropna=False))
```

```
   age  phd  educ   logwage  hs  somecoll  college  ma  govt  educ13  educ14
0   39    0    14  2.890372   0         1        0   0     0       0       1
1   48    0    12  3.292984   1         0        0   0     0       0       0
educ_level
1    14310
4    11265
2     8199
3     5180
5     4437
6     2013
Name: count, dtype: int64
```

1.1 1. Means of All Variables in the private and public sectors

Construct a table like the 1st table (page 7) in Lecture 3. Show the means of all the variables for workers in the private (govt=0) and public (govt=1) sectors, and the differences in these means.

[4]:
```
means = pay.groupby(['govt']).mean().reset_index(drop=False)
# print(means)
means_tab = means.transpose()
means_tab = means_tab.rename(columns={0:'private',1:'public'})
means_tab['diff'] = means_tab['public'] - means_tab['private']
print(means_tab)
# optional: produce LaTeX code, and customize... print(means_tab.to_latex())
```

```
              private     public       diff
govt         0.000000   1.000000   1.000000
age         40.147900  40.359072   0.211171
phd          0.040334   0.066852   0.026517
educ        14.212331  15.072982   0.860651
logwage      3.006674   3.218582   0.211908
hs           0.337146   0.191505  -0.145641
somecoll     0.293129   0.303313   0.010184
college      0.246310   0.258210   0.011900
ma           0.083080   0.180120   0.097039
educ13       0.180323   0.182017   0.001694
educ14       0.112806   0.121296   0.008490
educ_level   2.678858   3.253978   0.575120
```

1.2 2. Education and Wage Gaps

2. Construct a table like the 2nd table (page 8) in Lecture 3. Show the fractions of private and government workers at each education level, and the mean wages of the two groups in each education level, and the public-private wage gap at each level.

[5]:
```
# frac:
frac = pay.groupby(['govt'])['educ_level'].value_counts(normalize=True).
 ↪reset_index(drop=False)
# reshape to wide (each row corresponds to an educ_level)
frac = frac.pivot(index='educ_level',columns='govt',values='proportion')
frac = frac.rename(columns={0:'f_private',1:'f_public'})

# mean log wage:
wage = pay.groupby(['govt','educ_level'])['logwage'].mean().
 ↪reset_index(drop=False)
# reshape to wide (each row corresponds to an educ_level)
wage = wage.pivot(index=['educ_level'], columns=['govt'], values='logwage')
wage = wage.rename(columns = {0:'logwage_private',1:'logwage_public'} )

# merge together:
tab_by_educ = pd.merge(wage,frac,on=['educ_level'],how='outer')
tab_by_educ['govt_premium'] = tab_by_educ['logwage_public'] -␣
 ↪tab_by_educ['logwage_private']
print(tab_by_educ[['f_private','logwage_private','f_public','logwage_public','govt_premium']].
 ↪sort_values(['educ_level']))
```

```
govt         f_private  logwage_private  f_public  logwage_public  govt_premium
educ_level
1             0.337146         2.703243  0.191505        2.968421      0.265178
2             0.180323         2.871634  0.182017        3.121693      0.250059
3             0.112806         2.934910  0.121296        3.156861      0.221952
4             0.246310         3.256966  0.258210        3.273757      0.016791
5             0.083080         3.553914  0.180120        3.424559     -0.129355
6             0.040334         3.691762  0.066852        3.542903     -0.148859
```

1.3 3. Distribution across Education Categories, by Group

Using the notation of Lecture 3, let private sector workers be called "group a" (who are the reference group), let government workers be called "group b", and define the 6 category variables D_{gi} for each level of education g=1…6. Let $x_i' = (D_{1i}, D_{2i}, ...D_{6i})$. Find $\overline{x}^a$ and $\overline{x}^b$.

Answer: $\sum_{i\in a} D_{ki}$ is the *fraction* of workers in group "a" who have education level k. We have shown $\overline{x}_a$ (private) and $\overline{x}_b$ in the table above.

[6]:
```
x_a = tab_by_educ['f_private']
x_b = tab_by_educ['f_public']
print('group a (private sector)', np.array(x_a))
print('group b (public sector)', np.array(x_b))
```

```
group a (private sector) [0.33714627 0.18032319 0.11280575 0.24631027 0.08308043
0.04033409]
group b (public sector) [0.19150489 0.18201722 0.12129616 0.25821048 0.18011969
0.06685155]
```

1.3.1 (a) OLS: Log wage on education categories, by group

Fit an OLS regression model for wages for each group, and estimate the coefficients $\hat{\beta}^a$ and $\hat{\beta}^b$. Verify that these coefficients are the mean log wages of groups a and b, respectively, for each level of education.

[7]:
```
# model_a for Private Sector
model_a = smf.ols("logwage ~ 0 + C(educ_level)", data=pay.loc[pay['govt']==0]).
    ↪fit(cov_type='HC1')
print(model_a.summary())
beta_a = np.array(model_a.params)
print(beta_a)
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                logwage   R-squared:                       0.146
Model:                            OLS   Adj. R-squared:                  0.146
Method:                 Least Squares   F-statistic:                       nan
Date:                Wed, 10 Sep 2025   Prob (F-statistic):                nan
Time:                        16:59:42   Log-Likelihood:                -43182.
No. Observations:               38553   AIC:                         8.638e+04
Df Residuals:                   38547   BIC:                         8.643e+04
Df Model:                           5
Covariance Type:                  HC1
================================================================================
====
                       coef    std err          z      P>|z|      [0.025
0.975]
--------------------------------------------------------------------------------
----
C(educ_level)[1]     2.7032      0.006    447.322      0.000       2.691
2.715
C(educ_level)[2]     2.8716      0.009    336.752      0.000       2.855
2.888
C(educ_level)[3]     2.9349      0.011    278.631      0.000       2.914
2.956
C(educ_level)[4]     3.2570      0.008    399.254      0.000       3.241
3.273
C(educ_level)[5]     3.5539      0.014    254.940      0.000       3.527
3.581
C(educ_level)[6]     3.6918      0.024    151.691      0.000       3.644
3.739
==============================================================================
```

```
Omnibus:                     1513.746   Durbin-Watson:                   1.922
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2402.645
Skew:                          -0.358   Prob(JB):                         0.00
Kurtosis:                       3.991   Cond. No.                         2.89
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
[2.70324297 2.87163393 2.93490973 3.25696581 3.55391407 3.69176244]
```

[8]:
```python
# model_b for public sector
model_b = smf.ols("logwage ~ 0 + C(educ_level)", data=pay.loc[pay['govt']==1]).
 ↪fit(cov_type='HC1')
print(model_b.summary())
beta_b = np.array(model_b.params)
print(beta_b)
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                logwage   R-squared:                       0.104
Model:                            OLS   Adj. R-squared:                  0.104
Method:                 Least Squares   F-statistic:                       nan
Date:                Wed, 10 Sep 2025   Prob (F-statistic):                nan
Time:                        16:59:42   Log-Likelihood:                -5028.1
No. Observations:                6851   AIC:                         1.007e+04
Df Residuals:                    6845   BIC:                         1.011e+04
Df Model:                           5
Covariance Type:                  HC1
================================================================================
====
                     coef    std err          z      P>|z|      [0.025
0.975]
--------------------------------------------------------------------------------
----
C(educ_level)[1]     2.9684      0.014    207.072      0.000       2.940
2.997
C(educ_level)[2]     3.1217      0.015    212.322      0.000       3.093
3.151
C(educ_level)[3]     3.1569      0.017    187.290      0.000       3.124
3.190
C(educ_level)[4]     3.2738      0.012    274.493      0.000       3.250
3.297
C(educ_level)[5]     3.4246      0.014    252.181      0.000       3.398
3.451
C(educ_level)[6]     3.5429      0.025    142.121      0.000       3.494
3.592
==============================================================================
Omnibus:                      412.889   Durbin-Watson:                   1.822
```

```
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1838.187
Skew:                           0.005   Prob(JB):                         0.00
Kurtosis:                       5.538   Cond. No.                         1.97
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
[2.96842077 3.1216934  3.15686137 3.27375681 3.4245594  3.54290303]
```

I fitted regressions of logwage on education_level (categorical) without a constant. The coefficients $\hat{\beta}^a$ are mean log wage at each education level in group a (private sector), matched with the means in table 2 (tab_by_educ).

The coefficients $\hat{\beta}^b$ are mean log wage at each education level in group b (public sector), matched with the means in table 2 (tab_by_educ).

1.3.2 (b) Mean log wage as a weighted average

Using your estimates of $\overline{x}^a$, $\overline{x}^b$, $\hat{\beta}^a$ and $\hat{\beta}^b$, construct $\overline{y}^a = \sum_g \overline{p}^a_g \overline{y}^a_g = (\overline{x}^a)'\hat{\beta}^a$ and $\overline{y}^b = \sum_g \overline{p}^b_g \overline{y}^b_g = (\overline{x}^b)'\hat{\beta}^b$. Verify that these match the means of wages for groups a and b that you construct directly.

[9]:
```python
# Construct mean log wage in group a and b
y_a = np.sum(x_a * beta_a);
y_b = np.sum(x_b * beta_b)
print(f'y_a (private) = {y_a}'); print(f'y_b (public) = {y_b}')

# show they equal to the mean log wage by "govt"
print(pay.groupby(['govt'])['logwage'].mean())
```

```
y_a (private) = 3.0066738859268334
y_b (public) = 3.218581695120003
govt
0    3.006674
1    3.218582
Name: logwage, dtype: float64
```

1.3.3 (c) Counterfactual 1

Using your estimates of $\overline{x}^a$, and $\hat{\beta}^b$ construct

$$\overline{y}^b_{counterf} = \sum_g \overline{p}^a_g \overline{y}^b_g = (\overline{x}^a)'\hat{\beta}^b$$

Find the adjusted wage gap: $\overline{y}^b_{counterf} - \overline{y}^a$. How does this compare to the actual gap $\overline{y}^b - \overline{y}^a$?

[10]:
```python
y_b_counterf = np.sum(x_a * beta_b)
raw_gap = y_b- y_a
counterf_gap = y_b_counterf - y_a
print(f'Unadjusted gap = {raw_gap}')
print(f'Holding fixed the educ distribution, the remaining gap =␣
  ↪{counterf_gap}')
```

```
Unadjusted gap = 0.21190780919316943
Holding fixed the educ distribution, the remaining gap = 0.14691749105708007
```

1.3.4 (d) Counterfactual 2

Construct the weight $w_g = N_g^a/N_g^b$ for education categories $g = 1...6$. Use this to construct $\overline{y}^b_{counterf2} = \frac{\sum_{i\in b} w_g y_i}{\sum_{i\in b} w_g}$ and verify that:

$$\overline{y}^b_{counterf} = \sum_g \overline{p}^a_g \overline{y}^b_g = (\overline{x}^a)'\widehat{\beta}^b = \frac{\sum_{i\in b} w_g y_i}{\sum_{i\in b} w_g} = \overline{y}^b_{counterf2}$$

.

[11]:
```python
# construct weights:
N_a = (pay['govt']==0).sum(); N_b = (pay['govt']==1).sum()
# note: N_g(a) = N_a* x_a, N_g(b) = N_b*x_b ~ num. of workers at each education␣
↪level, in private and public sectors respectively
w_g = (x_a * N_a)/(x_b*N_b)
print(w_g)

# merge the weights with the dataframe "pay" by educ_level:
w_g = pd.DataFrame(w_g).reset_index(drop=False); w_g.columns␣
↪=['educ_level','weight']
pay = pd.merge(pay, w_g, on=['educ_level'],how='left')

# counterfactual: if in group a (private), weight=1
print(pay.groupby(['govt'])['weight'].sum()) # note
# pay.loc[pay['govt']==0,'weight']=1 # no need to weight workers in group
```

```
educ_level
1    9.907012
2    5.574980
3    5.233454
4    5.368005
5    2.595624
6    3.395197
dtype: float64
govt
0    254856.780292
1     38553.000000
Name: weight, dtype: float64
```

Note there is no need to weight people in group a (private sector). The sum of weights for workers in group b (public) sum up to N_a = num. workers in group a:

$$\sum_{i\in b} w_g = \sum_g N_g^b \times (N_g^a/N_g^b) = \sum_g N_g^a = N_a$$

[19]:
```python
assert abs(pay.loc[pay['govt']==1,'weight'].sum()-N_a)<1e-5
```

construct the 2nd counterfactual with weight w_g on workers in group b with education level = g:

$$\bar{y}^b_{counterf2} = \frac{\sum_{i \in b} w_g y_i}{\sum_{i \in b} w_g}$$

[18]:
```
pay['wtd_logwage'] = pay['weight']* pay['logwage']
y_b_counterf2 = pay.loc[pay['govt']==1,'wtd_logwage'].sum() / N_a

print(y_b_counterf2, y_b_counterf)
# verify they are equal (allowing for rounding error):
assert abs(y_b_counterf2-y_b_counterf)<1e-5
```

```
3.1535913769839197 3.1535913769839135
```